

Historical Quantitative Reasoning on the Web

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Abstract. The Semantic Web is an extension of the Web through standards by the World Wide Web Consortium (W3C) [3]. These standards promote common data formats and exchange protocols on the Web, most fundamentally the Resource Description Framework (RDF). Its ultimate goal is to make the Web a suitable data space for humans (using the paradigm of Linked Documents) as well as for machines (using the paradigm of Linked Data). The latter has experimented an enormous growth in the last years, mostly due to the adoption of Linked Data publishing practices by institutions, governments and users, giving birth to the Linked Open Data cloud: a Web-graph of 100 billion interconnected facts and schemas (also called ontologies) [2]. Many domains have published their datasets as Linked Open Data, including History [4,5], converting them to RDF and linking them to related historical datasets and concepts on the Web. One of the fundamental problems of historical research are so-called gaps in historical evidence: the non-existence of relevant historical data for a particular matter. The very exercise of historical research has a primary focus on filling these gaps, generating the missing knowledge (to a level of certainty) using diverse methods. When raw historical datasets are published on the Semantic Web as Linked Data, these gaps still exist [1]. In this paper, we investigate whether standard Semantic Web reasoning facilities [9] are useful to fill these gaps of historical knowledge. We find that standard entailment regimes used on the Web are insufficient to achieve satisfactory results. With the assumption that historical reasoning is based on a non-standard entailment regime, we study the way in which social historians derive new knowledge from historical quantitative sources to fill missing gaps [1,6]. We mimic their behavior in a Semantic Web setting, by adapting existing reasoning technology [7] to historical research, and checking whether automated quantitative reasoning can entail the same historical conclusions. We explore whether these automated reasoning mechanisms can be generalized to be applied to other domains [8], within and out of History.

Keywords: Reasoning, Semantic Web, Statistics, Social and Economic History

References

1. Ashkpour, A., Meroño-Peñuela, A., Mandemakers, K.: The Dutch Historical Censuses: Harmonization and RDF. *Historical Methods: A Journal of Quantitative and Interdisciplinary History* 48 (2015)
2. Beek, W., Rietveld, L., Bazoobandi, H.R., Wielemaker, J., Schlobach, S.: LOD Laundromat: A Uniform Way of Publishing Other People’s Dirty Data. In: *The Semantic Web – ISWC 2014* (2014)
3. Berners-Lee, T., Hendler, J., Lassila, O.: The Semantic Web. *Scientific American* 284(5), 34–43 (2001)
4. Burrows, T.: A data-centred ‘virtual laboratory’ for the humanities: Designing the Australian Humanities Networked Infrastructure (HuNI) service. *Literary and Linguistic Computing: The Journal of Digital Scholarship in the Humanities* 28(4), 576–581
5. Meroño-Peñuela, A., Ashkpour, A., van Erp, M., Mandemakers, K., Breure, L., Scharnhorst, A., Schlobach, S., van Harmelen, F.: Semantic Technologies for Historical Research: A Survey. *Semantic Web – Interoperability, Usability, Applicability* 6(6), 539–564 (2015)
6. Meroño-Peñuela, A., Ashkpour, A., Guéret, C., Schlobach, S.: CEDAR: The Dutch Historical Censuses as Linked Open Data. *Semantic Web – Interoperability, Usability, Applicability* (2015), In press
7. Stringer, B., Meroño-Peñuela, A., Loizou, A., Abeln, S., Heringa, J.: SCRY: Enabling quantitative reasoning in SPARQL queries. In: *Semantic Web applications and tools for life sciences (SWAT4LS 2015)*, December 7-10 Cambridge, England (2015) (2015)
8. Stringer, B., Meroño-Peñuela, A., Loizou, A., Abeln, S., Heringa, J.: To SCRY Linked Data: Extending SPARQL the Easy Way. In: *Proceedings of the Diversity++ workshop. International Semantic Web Conference (ISWC 2015)* (2015)
9. The World Wide Web Consortium (W3C): OWL Web Ontology Language Overview. <http://www.w3.org/TR/owl-features/>